

Telephony: Asterisk, SIP, and a Real Phone Number

Optional. Not required for the prototype, and deliberately not part of first run — it adds meaningful setup complexity for no demo benefit.

Be clear about cost

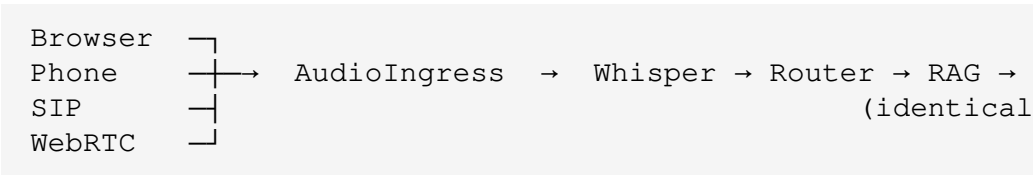
A real phone number cannot be free. Software is free; carrier connectivity is not. Anyone claiming otherwise is describing a trial credit.

Option	Software cost	Carrier cost	Setup
A. Browser demo	\$0	\$0	✓ already built
B. Local Asterisk	\$0	\$0 (SIP softphone only, no PSTN)	~2 hours
C. Real phone number	\$0	~\$1–2/month + ~\$0.013/min	~1 day

Option B gives you a real PBX and real SIP calls between softphones on your network — genuinely useful for testing telephony behavior — but it **cannot reach an actual phone number** without a trunk from a provider.

Why the AI layer needs no changes

`backend/app/api/voice.py` accepts audio bytes and returns audio bytes. It does not know or care about the source:



`channel` is recorded for analytics only. Adding telephony means writing a bridge that posts audio to `/api/voice/turn` and plays back the response — the reasoning pipeline is untouched.

Option B — Local Asterisk

Install

```
brew install asterisk          # macOS
sudo apt install asterisk      # Debian/Ubuntu
```

Minimal configuration

`pjsip.conf` — one softphone extension:

```
[transport-udp]
type = transport
protocol = udp
bind = 0.0.0.0:5060

[6001]
type = endpoint
context = garden-city
disallow = all
; ulaw is 8kHz mono - matches what Whisper expects and av
```

```

allow = ulaw
auth = 6001-auth
aors = 6001

[6001-auth]
type = auth
auth_type = userpass
username = 6001
; Local testing only. Never reuse this anywhere real.
password = change_this_local_only

[6001]
type = aor
max_contacts = 1

```

`extensions.conf` — route calls to the AI bridge:

```

[garden-city]
; Dial 1000 to reach the assistant
exten => 1000,1,Answer()
    same =>          n,Wait(1)
    same =>          n,Playback(hello-world)
    same =>          n,AGI(garden_city_ai.agi)
    same =>          n,Hangup()

; Simulated department transfers become real ones here
exten => 2001,1,Dial(PJSIP/public_works,30)
exten => 2002,1,Dial(PJSIP/recreation,30)
exten => 2003,1,Dial(PJSIP/building,30)
exten => 2004,1,Dial(PJSIP/village_clerk,30)

```

Connect a softphone

Install [Zoiper](#) or [Linphone](#), register as extension 6001 against your machine's IP, and dial `1000`.

The bridge

You need an AGI or ARI script that:

1. Records caller audio until silence (Asterisk's `Record()` with a silence timeout, or ARI snoop for streaming)
2. POSTs the audio to `http://127.0.0.1:8000/api/voice/turn` with `channel=phone`
3. Fetches speech from `/api/voice/synthesize`
4. Plays it back
5. On an `escalate` action, dials the department extension instead

This bridge is not included. It is a genuine piece of work — a few hundred lines with real edge cases around barge-in, DTMF, and call teardown — and it would sit unused in a browser-only demo.

Recommended approach: **ARI (Asterisk REST Interface) over AGI.** ARI supports streaming media and external media channels, which makes interruption handling far cleaner than AGI's turn-based model.

Option C — A real phone number

Path 1: Twilio (simplest)

1. Buy a number (~\$1.15/month)
2. Point its Voice webhook at your backend
3. Use Twilio Media Streams to send audio over a WebSocket
4. Bridge that WebSocket to `/api/voice/turn`

Cost: number + ~\$0.0085/min inbound. A pilot at 100 calls/day averaging 3 minutes is roughly **\$80/month**.

Your backend needs a public HTTPS URL — `cloudflared tunnel` works for testing; production needs real hosting (see [PRODUCTION ROADMAP.md](#)).

Path 2: SIP trunk into Asterisk

Providers: Flowroute, Telnyx, VoIP.ms, Bandwidth.

Usually cheaper per minute and keeps media on Village infrastructure — which often matters more than cost for a municipality, since resident audio never transits a third-party API.

```
[trunk-provider]
type = registration
transport = transport-udp
outbound_auth = trunk-auth
server_uri = sip:sip.provider.example
client_uri = sip:VILLAGE_DID@sip.provider.example
```

Credentials belong in a secret store, not in a committed config file.

Telephony-specific requirements

Things that don't matter in a browser and matter a great deal on a phone:

Concern	Why it changes
Audio format	Phone audio is 8 kHz μ -law. Whisper wants 16 kHz. Resample, don't just reinterpret.
Barge-in	Callers interrupt constantly. Needs echo cancellation so the assistant doesn't hear itself.
Latency budget	Silence over 2s reads as a dropped call. Current warm turn is ~3s — acceptable, but the greeting should start immediately.
DTMF fallback	Offer "press 1 for..." for callers who prefer keypads or have speech differences.
Call teardown	Hangups mid-turn must clean up sessions and not orphan work.

Recording disclosure	Legally required in most jurisdictions. Must be spoken at call start.
911	The assistant must never attempt to handle an emergency. The policy override exists; on a phone line, confirm the carrier path to emergency services is unaffected.
Accessibility	Callers using relay services, TTY, or with atypical speech must reach a human easily.

Recommended sequence

1. **Browser demo** (done) — proves the AI works
2. **Local Asterisk + softphone** — proves the telephony bridge works, \$0
3. **SIP trunk or Twilio number, staff-only** — real phone, real handsets, no residents
4. **Pilot on one department** — publish the number narrowly, keep human fallback
5. **Expand** only after measured accuracy holds

Do not skip step 2. Debugging an audio bridge and a carrier connection at the same time is much harder than debugging them separately.